

DINOJAN VIVEKANANTHARASA

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PROFESSIONAL SUMMARY

A Computer Engineering undergraduate with deep expertise in **AI systems architecture and applied machine learning**. Experienced in designing end-to-end AI/ML pipelines for security, threat detection, data retrieval (RAG systems), and LLM-based automation. Proven ability to ship research-grade systems with rigorous evaluation and real-world validation.

EDUCATION

BSc.Eng (Honours) in Computer Engineering

Mar 2022 – Present

Faculty of Engineering, University of Ruhuna, Galle, Sri Lanka

- **Soft Skill Development** (2022–2024)

GCE Advanced Level

Aug 2018

V/Vavuniya Tamil Madhya Maha Vidyalayam

PROFESSIONAL EXPERIENCE

Full Stack Developer

December 2025 – May 2026

Yarl Ventures (PVT) Ltd, Jaffna

Full Stack Developer Intern

May 2025 – November 2025

Yarl Ventures (PVT) Ltd, Jaffna

TECHNICAL SKILLS

Programming: Python, JavaScript, Node.js, SQL, C++, Java

AI/ML: LLM fine-tuning (QLoRA), Retrieval-Augmented Generation (RAG), Transfer Learning, CNN, TensorFlow, PyTorch, Scikit-learn

Tools: FastAPI, Streamlit, ChromaDB, Sentence-Transformers, OpenAI/Groq API, Docker, AWS, Hyperledger Fabric

NON-TECH ABILITIES

Skills: Time Management, Goal-Oriented, Teamwork, Problem Solving

Languages: English (Fluent), Tamil (Native), Sinhala (Basic)

PROJECTS

TRACES – Post-Quantum Secure Vehicular Network Defense

Dec 2025 – Present

Python, NS-3, Qwen2.5-7B, Hyperledger Fabric

- Designed a three-stage LLM-based anomaly detection cascade and zero-trust blockchain governance for vehicular networks, validated on 200-vehicle simulation.

AI-Powered Financial Document Q&A System (RAG)

Jan 2026 – May 2026

Python, FastAPI, Streamlit, ChromaDB

- Built end-to-end RAG pipeline for financial documents achieving Hit@1 retrieval = 1.000, recovered 84% gold-answer keywords vs under 1% for ungrounded LLM; deployed live on Streamlit.

AutoGradePro – AI-Powered Grading System

Jun 2024 – May 2025

Next.js, TypeScript, Ollama

- Built Next.js frontend for AI-based grading platform with Ollama semantic matching, supporting TXT, DOCX, and PDF formats with role-based access.

ECG Image Classification

Dec 2024 – Apr 2025

Python, TensorFlow, CNN

- Developed CNN model for ECG classification on MIT-BIH and PTB datasets with transfer learning; built end-to-end pipeline from preprocessing to evaluation.

Personal AI Assistant Web App

Apr 2025 – May 2025

React.js, Node.js, OpenAI API

- Built full-stack conversational AI assistant with OpenAI integration, JWT auth, persistent chat history, and adaptive response behavior.

Room Occupancy Detection

Jun 2024 – Nov 2024

Python, Scikit-learn

- Built ML models for IoT-based room occupancy classification for smart buildings; handled full pipeline from preprocessing to evaluation.

PUBLICATIONS

G. S. J. Venujan, M. F. A. Munsif, **V. Dinojan**, T. Jegan, and G. G. N. Sandamali, “AutoGradePro: An Automated Grading System for Short-Answer Assessments Using Rule-Based Methods and Large Language Models,” in *Proceedings of Academic Sessions 2026*, University of Ruhuna, Sri Lanka, Mar. 2026, Abstract ABS-064. ISSN 2362-0412.

VOLUNTEER EXPERIENCE

Demonstrator – ReXtro 2025

Dec 2025

Faculty of Engineering, University of Ruhuna

Member

Apr 2024 – Present

IESL, Galle

Member

Sep 2023 – Dec 2024

IEEE Student Branch, Ruhuna, Galle

Treasurer

May 2021 – April 2024

All University Students' Development Association, Vavuniya

Senior Prefect

July 2017 – July 2018

V/V.T.M.M.V, Vavuniya

ACHIEVEMENTS

Participated in **IEEE Xtreme 18.0 (2024)**, a global 24-hour algorithmic coding contest (3-member team)

Competed in **IEEE INSL Southern Provincial Competition**, 30th October 2024

Took part in the **Galle Division FIDE 100 Commemorative Chess Event** (Open Category), 2024

REFEREES

- **Dr. Kushan Sudheera**

Senior Lecturer

Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna

Contact: +94 719693164 Email: kushan@eie.ruh.ac.lk

- **Dr. Nadeesha Sandamali**

Senior Lecturer

Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna

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